THE CANONICAL DECOMPOSITION OF A MODULE-VALUED FUNCTOR ON SQUARE-ZERO EXTENSIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof writes TF(N, B) once where the definition in item (1) gives TF(B, N); this is preserved verbatim and recorded here rather than corrected.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The construction of the square-zero extension B[N] and the functoriality of the pair (B, N). Same chapter, the discussion of square-zero extensions preceding the lemma. Supplies the algebras B[N] and the morphisms of pairs used throughout.

Lemma 0.1. *Let A be a ring. Let F be a module valued functor. For every $B \in \mathrm{Ob}(Alg_A)$ and B-module N there is a canonical decomposition*

$$F(B[N]) = F(B) \oplus TF(B, N)$$

characterized by the following properties

(1) $TF(B, N) = \mathrm{Ker}(F(B[N]) \to F(B))$,
(2) *there is a B-module structure $TF(B, N)$ compatible with $B[N]$-module structure on $F(B[N])$,*
(3) *TF is a functor from the category of pairs (B, N),*
(4) *there are canonical maps $N \otimes_B F(B) \to TF(B, N)$ inducing a transformation between functors defined on the category of pairs (B, N),*
(5) *$TF(B, 0) = 0$ and the map $TF(B, N) \to TF(B, N')$ is zero when $N \to N'$ is the zero map.*

Proof. Since $B \to B[N] \to B$ is the identity we see that $F(B) \to F(B[N])$ is a direct summand whose complement is $TF(N, B)$ as defined in (1). This construction is functorial in the pair (B, N) simply because given a morphism of pairs $(B, N) \to$

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(B', N') we obtain a commutative diagram

$$\begin{array}{ccccc} B' & \longrightarrow & B'[N'] & \longrightarrow & B' \\ \uparrow & & \uparrow & & \uparrow \\ B & \longrightarrow & B[N] & \longrightarrow & B \end{array}$$

in Alg_A. The B-module structure comes from the $B[N]$-module structure and the ring map $B \to B[N]$. The map in (4) is the composition

$$N \otimes_B F(B) \longrightarrow B[N] \otimes_{B[N]} F(B[N]) \longrightarrow F(B[N])$$

whose image is contained in $TF(B, N)$. (The first arrow uses the inclusions $N \to B[N]$ and $F(B) \to F(B[N])$ and the second arrow is the multiplication map.) If $N = 0$, then $B = B[N]$ hence $TF(B, 0) = 0$. If $N \to N'$ is zero then it factors as $N \to 0 \to N'$ hence the induced map is zero since $TF(B, 0) = 0$. $\square$